FORMULA SHEET

QOIDA · Bir xil asosli daraja qoidalari

Qoida	Misol
$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^{3+4} = 2^7$
$(a^m)^n = a^{mn}$	$(2^7)^2 = 2^{7\times 2} = 2^{14}$
$\frac{a^m}{a^n} = a^{m-n}$	$\frac{2^{14}}{2^5} = 2^{14-5} = 2^9$

QOIDA · Nol va manfiy daraja

Nolmas a uchun: $a^0 = 1$. Manfiy ko'rsatkich – teskari sonni bildiradi, asosni manfiy qilmaydi:

$$5^0 + 2^{-3} \times 4^2$$

← 1. $a^0=1$, $a^{-n}=\frac{1}{a^n}$

$$\downarrow$$

$$1 + \frac{1}{8} \times 16$$

← 2. ko'paytiring

$$\downarrow$$

$$1 + 2 \qquad = 3$$

QOIDA · Kvadrat ildizni soddalashtirish

Eng katta to'liq kvadrat ko'paytuvchini toping, so'ng ajrating: $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$.

$$\sqrt{72}$$

← 1. $72=36\times 2$, 36 – to'liq kvadrat

$$\downarrow$$

$$\sqrt{36 \times 2} = \sqrt{36} \times \sqrt{2}$$

← 2. ajrating

$$\downarrow$$

$$= 6\sqrt{2}$$

Eslatma: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ *– masalan* $\sqrt{16+9} = 5$, *biroq* $\sqrt{16} + \sqrt{9} = 7$.